

# 23CSE312 DISTRIBUTED SYSTEMS

S6 B.Tech CSE

Department of Computer Science and Engineering  
Amrita School of Computing, Amritapuri Campus

## CASE STUDY

### Apache Kafka in the World Wide Web as a Distributed System

Group No : 19

Team Members

| Roll No          | Student Name   |
|------------------|----------------|
| AM.SC.U4CSE23210 | Harsha Vardhan |
| AM.SC.U4CSE23219 | Tejaswini      |
| AM.SC.U4CSE23253 | Gowtham        |
| AM.SC.U4CSE23272 | Chetan Krishna |



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## Abstract

The Apache Kafka is a highly scalable, fault-tolerant, distributed system, which is capable of streaming and processing real-time data across the whole World Wide Web. This particular case study focuses on the Kafka architecture, key components, and its function as a supporting framework for the modern web-scale applications. Initially developed to handle with high throughput data pipelines, Kafka has evolved into a powerful event streaming platform that organizations utilize for managing their continuous data flows efficiently.

It analyzes Kafka's methods for distribution through its partitioned log model, replication processes, and decentralized broker architecture. The study also breaks down the central components, such as producers, consumers, topics, and brokers, to see how data is pushed in, held, and taken out within the distributed environment. Besides, it refers to the ways in which Kafka is capable of providing durability, scalability, and fault tolerance, all the while retaining low latency.

Examples of the implementation of Kafka in the real world in different areas, such as activity tracking, log aggregation, and stream processing are mentioned here to illustrate the role of Kafka in the web-based distributed systems. Furthermore, the study touches on topics like data consistency, ordering guarantees, and complexity of the system.

In conclusion, this case study gives an overview of the mechanics of Apache Kafka as a distributed system on the Internet, highlighting the role it has played in the emergence of real-time, data-driven applications.

**Keywords:** Apache Kafka, Distributed Systems, Event Streaming, Message Queuing, Data Pipelines, Real-Time Processing, Fault Tolerance, Scalability, Microservices Architecture, Data Replication

# 1. Introduction

## 1.1 Background

Apache Kafka started as a project at LinkedIn to manage the data pipelines of their business and to enable real-time event streaming. Such data streams can be user activity on the website, logs, transactional data, sensor data, etc.

Traditional systems with a single central processing instance are not able to efficiently handle such a big volume of data. Distributed systems worked out that problem by allowing data processing to be conducted on several nodes and thus making the system not only scalable but also fault tolerant.

Kafka plays well with others in these cases. First of all, it is a distributed publish, subscribe messaging system based on a partitioned log model. It allows producers to continue sending messages to a topic, and consumers to read from the topic independently. It is well suited for real-time web applications.

Overtime, it became a key player in data architectures. Kafka docks with Spark and Flink well to provide stream processing and analytics. In fact, it is by far one of the foundational tools in distributed web systems, able to handle big volumes of data with very low latency, durability, and horizontal scalability.

## 1.2 Problem Statement

The rapid growth of web-based applications, has been a rising need for systems capable of processing and transferring the data in real time while simultaneously ensuring reliability and scalability. Old messaging systems sometimes give up in terms of throughput, fault tolerance and the capability of handling distributed workloads. Such drawbacks make the systems work slower in scenarios where uninterrupted data stream is most important, e.g., real-time analyses, monitoring and event-driven architectures.

The motivation behind this case study is to analyze how a distributed platform such as Apache Kafka manages to tackle these issues. In particular, the case study is intended to explore the ways in which Kafka guarantees high availability, data persistence and scalability even when it is running in a decentralized manner.

By analyzing Kafka's architecture and operational principles, this report aims to demonstrate its effectiveness in addressing real-world distributed system problems and its importance in enabling modern, data-intensive web applications.

# 2. System Design and Architecture

## 2.1 System Architecture Overview

Apache Kafka uses a distributed, publish, subscribe framework that can handle high-throughput, real-time data streams. Where Kafka steps from is a collection of servers

(brokers), working as a team to store and manage data streams in a fault-tolerant fashion.

Kafka's data is organized into topics and then partitioned among these topics. A partition is simply a continuous sequence of records that are never modified, and instead, new records are always appended to them. To allow for concurrent processing as well as increased scalability, these partitions are distributed on different brokers. Replication or keeping copies of each partition on different brokers is the main feature of fault tolerance. Among these replicas, one is designated as the leader while the rest are followers.

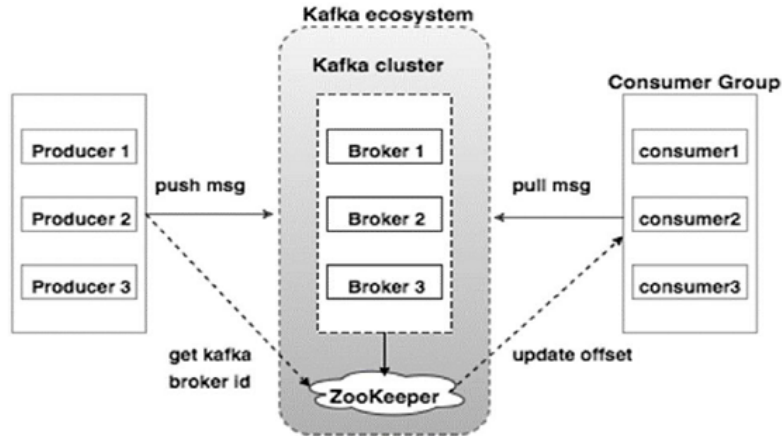


Figure 1: Overview of Apache Kafka architecture illustrating the flow of data from producers to consumers through brokers in a distributed system.

Probably the most remarkable feature of Kafka's design is decentralization - it's a system that doesn't have a single point of failure. Producers send data to topics without having to know how the system works internally, and consumers on the other hand fetch data at their own pace and independently. This kind of separation between data producers and consumers is what makes Kafka so flexible and scalable for web systems of big scale.

## 2.2 Major Components and Their Roles

Kafka's architectural core include various components that collectively create a dependable, scalable and distributed data streaming platform.

- **Producers:** These are applications or systems that create data that is then stored in Kafka topic. In fact, producers are responsible for identifying which partition the data should be directed to; that is how load balancing and data spreading are more effective.
- **Consumers:** These are the individuals who subscribe to the topics and fetch the data. Further, to manage bigger workloads, Consumers can divide and conquer the partitions concurrently.

- **Brokers:** Kafka Brokers are the servers that not only save the data but also serve the clients' request. A Kafka cluster is made up of several brokers. They are the ones who manage the partitions and make sure that the data is always available.
- **Topics:** Methods that data can be arranged. In fact, each individual topic may be broken down into multiple partitions for distributed storage and processing.
- **Partitions:** The subdivisions of topics that allow multiple lines of work to be carried out simultaneously. In fact, each partition consists of an ordered list of messages, so that the data within that partition is consistent.
- **Replication Mechanism:** The replication mechanism is what allows a system running to be maintained even in the face of failures, by having multiple copies of partitions spread over different brokers. So, if a broker fails, the replica will automatically take the leadership role.
- **ZooKeeper / KRaft (Metadata Management):** Before, the Apache ZooKeeper was employed for storing the cluster's metadata and for cluster coordination. The recent Kafka versions that support KRaft mode do not need ZooKeeper and hence they simplify the architecture.

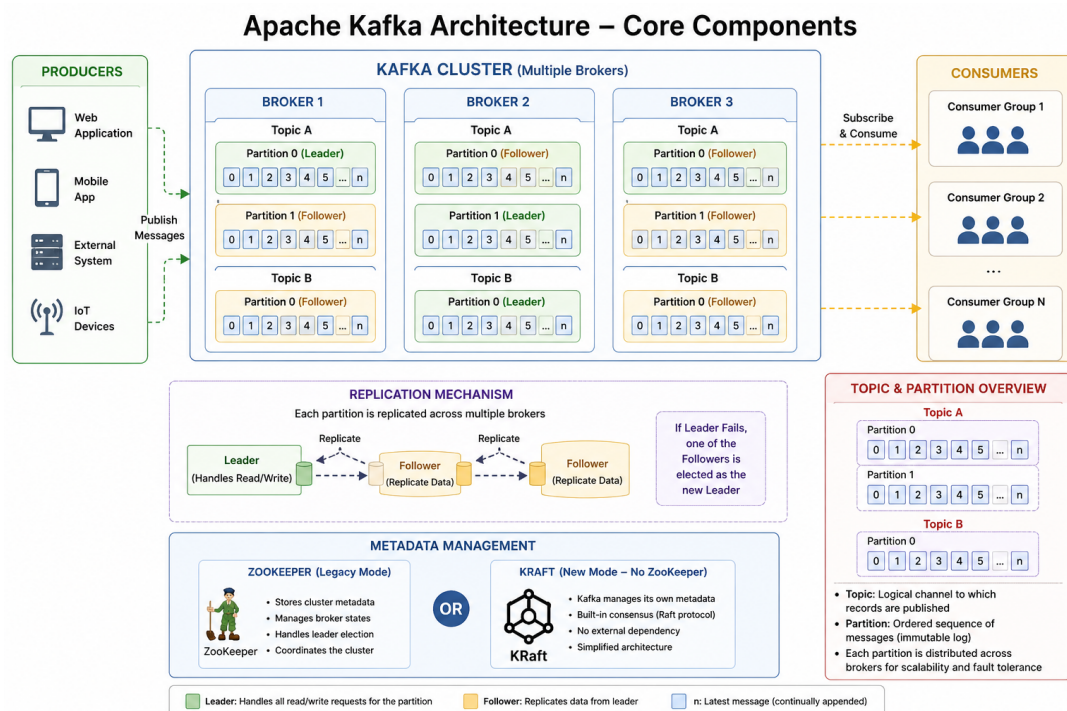


Figure 2: Apache Kafka Architecture Illustrating Core Components Including Producers, Consumers, Brokers, Topics, Partitions, Replication, and Metadata Management

### 3. Distributed Systems Principles

#### 3.1 Distributed Systems Concepts Applied

Apache Kafka incorporates several fundamental distributed systems concepts:

- **Event Ordering & Logical Time:** Kafka guarantees strict ordering within a partition. Each message is assigned an offset, which acts as a logical timestamp, ensuring deterministic event sequencing.
- **Leader Election:** Each partition has a leader broker responsible for read/write operations, while followers replicate data. Leader election ensures availability in case of broker failure.
- **Concurrency Control:** Multiple producers and consumers operate concurrently. Kafka handles this using partitioning and consumer groups, enabling parallel data processing without conflicts.
- **Consensus Mechanism:** Kafka ensures consistency among replicas using protocols similar to consensus systems. Modern Kafka (KRaft mode) removes dependency on Apache ZooKeeper and internally manages metadata consensus.
- **Consistency Model:** Kafka provides strong consistency within partitions and configurable durability using acknowledgment settings (e.g., acks=all).

#### 3.2 Communication and Coordination Mechanisms

Kafka employs quite effective communication and coordination mechanisms:

- **Publish-Subscribe Pattern:** In this mode, manufacturers deliver their messages to 'topics', and consumers independently subscribe to these topics, which established a concept of a loosely coupled system.
- **Pull-Driven Consumption:** Consumers can request the data from the brokers at their convenience, which helps in better scalability and avoiding the problem of overwhelm.
- **Consumer Groups and Rebalancing:** Kafka distributes partitions automatically and evenly among the consumers of a group. When a consumer is added or removed rebalancing guarantees redistribution of the workload.
- **Broker Communication:** Brokers interact with each other with the purposes of replication and metadata updating, which leads to the maintenance of consistency over the cluster.

### 3.3 Consistency, Replication, and Fault Tolerance

- **Replication Strategy:** A partition in Kafka is replicated to several brokers. The leader version of partition services the requests, and followers copy data either asynchronously or synchronously.
- **In-Sync Replicas (ISR):** Out of all the replicas Kafka only considers those who are completely synchronized as a set of in-sync replicas. These members alone are considered as potential leaders during election, which keeps data consistent.
- **Fault Tolerance:** In the event of a broker failure, a replica from the ISR can be promoted to leader, thus providing continuous service
- **Durability:** Apart from storing the data on the disk, one can tune Kafka to even stronger guarantees through replication and acknowledgment configurations.

### 3.4 Scalability and Performance Features

- **Horizontal Scalability:** When we want to scale Kafka, we can add more brokers and partitions to split the workload across the cluster.
- **High Throughput:** Kafka can process millions of messages per second thanks to its use of sequential disk writes and zero-copy transfer methods.
- **Partition-Level Parallelism:** To get better performance, different partitions can be processed by separate consumers at the same time.
- **Batch Processing & Compression:** By batching and compressing, Kafka is able to get the most out of network and storage.
- **Low Latency:** Kafka is built with the real-time applications in mind and is very efficient in message delivery with no or very little delay.

### 3.5 Security and Reliability Considerations

- **Authentication & Authorization:** Kafka can be set up to use SSL/TLS for encrypting data transmission and SASL for client authentication to ensure secure communication with brokers.
- **Access Control:** Role-based access control (ACLs) is employed to limit which users can perform read/write operations on specific topics.
- **Data Reliability:** The use of replication along with acknowledgments keeps data safe even in case of failures.
- **Monitoring & Logging:** Kafka supports integration with various monitoring tools for keeping an eye on broker health, throughput, and failures.

- **Exactly-Once Semantics (EOS):** Kafka provides guarantees to avoid duplicate processing in stream processing scenarios, enhancing reliability in critical applications.

## 4. Applications and Evaluation

### 4.1 Real-World Use Cases / Deployment Scenarios

Apache Kafka is widely used across industries for handling real-time data streams:

- **Activity Tracking & User Analytics:** Companies like LinkedIn use Kafka to track user interactions (clicks, searches, views) in real time for analytics and recommendations.
- **Log Aggregation:** Kafka collects logs from multiple servers and centralizes them for monitoring and debugging in distributed environments. Stream Processing Pipelines: Integrated with Apache Spark and Apache Flink for real-time data transformation, filtering, and analytics.

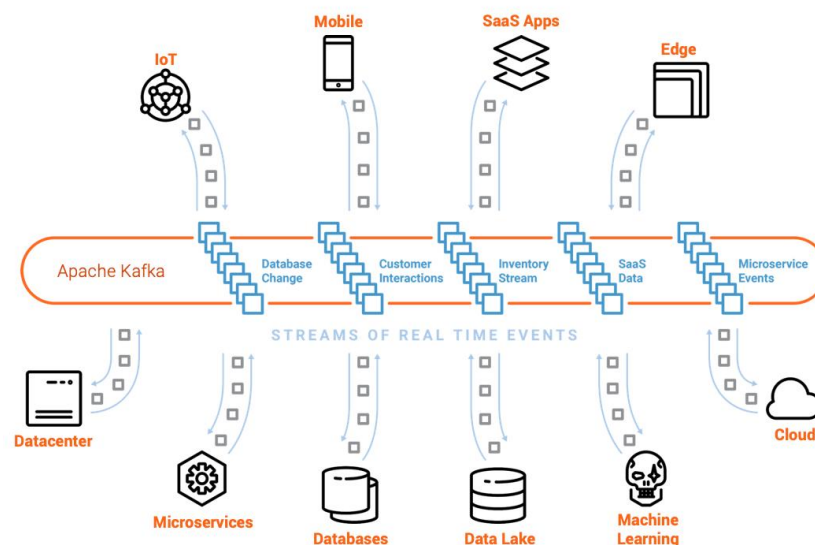


Figure 3: Apache Kafka enabling real-time event streaming across diverse sources such as IoT, mobile applications, SaaS platforms, and data systems, supporting analytics, microservices, and machine learning pipelines.

- **Event-Driven Microservices:** Acts as a backbone for communication between microservices, enabling asynchronous and decoupled architectures.
- **Financial Transactions & Monitoring:** Used in banking and fintech systems for fraud detection and real-time transaction processing.



## 4.2 Advantages of the System

- **High Throughput & Low Latency:** Efficient handling of large-scale data streams in real time.
- **Scalability:** Easily scales horizontally by adding brokers and partitions.
- **Fault Tolerance:** Replication and leader election ensure high availability.
- **Decoupling of Systems:** Producers and consumers operate independently, improving system flexibility.
- **Durability:** Persistent storage ensures data is not lost even during failures.

## 4.3 Limitations

- **Operational Complexity:** Managing clusters, partitions, and configurations can be complex for beginners.
- **Message Ordering Limitation:** Ordering is guaranteed only within a partition, not across the entire topic.
- **Latency in Some Scenarios:** Not ideal for ultra-low latency use cases compared to in-memory systems.
- **Data Management Overhead:** Requires careful handling of retention policies and storage management.
- **Learning Curve:** Understanding internal concepts like offsets, partitions, and replication requires effort.

## 4.4 Comparison with Similar Technologies

- **RabbitMQ:** Focuses on traditional message queuing with complex routing. Kafka is better for high-throughput streaming, while RabbitMQ excels in task-based messaging.
- **Apache Pulsar:** Similar to Kafka but separates storage and compute layers. Pulsar offers multi-tenancy and geo-replication more natively.
- **Amazon Kinesis:** A managed streaming service by Amazon Web Services. Easier to set up but less flexible compared to Kafka's open-source ecosystem.
- **ActiveMQ:** A traditional messaging system supporting multiple protocols. Kafka outperforms it in scalability and stream processing use cases.

## 4.5 Recent Improvements

- **KRaft Mode (Kafka without ZooKeeper):** Newer versions of Apache Kafka eliminate dependency on Apache ZooKeeper, simplifying deployment and improving scalability.
- **Cloud-Native Kafka:** Adoption of managed Kafka services like Confluent Cloud and cloud integrations is increasing.
- **Event Streaming as a Platform:** Kafka is evolving beyond messaging into a full event streaming ecosystem with tools for processing, connectors, and governance.
- **Integration with AI/ML Pipelines:** Kafka is increasingly used to stream data into machine learning systems for real-time inference and analytics.
- **Enhanced Exactly-Once Semantics:** Improvements in transactional processing ensure higher reliability in critical applications.

## 5. Critical Discussion

### 5.1 Critical Analysis and Key Insights

Apache Kafka aligns really well to the main distributed systems principles of scalability, fault tolerance, and high-throughput data handling. Partitioned log model is one of its biggest strengths and that means that it is possible to scale it horizontally and run in parallel, both are must-haves for nowadays web-scale systems. Producers and consumers are two totally independent components, which makes architecture very flexible, consequently, Kafka is a perfect fit for the microservices and event-driven systems.

That being said, there are also some design points of Kafka that result in the issuance of the balance between advantages and disadvantages: For instance, even though it is ordered within partitions, the global ordering across topics is not guaranteed. This can be an issue in the applications that require very strict sequencing. Moreover, its partitioning-based operation requires a lot of consideration since a partitioning strategy that is not up to the mark might lead to an unbalanced load and, therefore, the emergence of bottlenecks in performance.

What is more, not only Kafka cluster setup and configuration tuning (e.g., replication factor, retention policies) but also failure handling is a highly challenging task that requires a great deal of expertise. Nonetheless, features such as the KRaft mode has made deployment a bit easier but still to manage large-scale Kafka systems a good monitoring and operational discipline are a must.

Another important point to remember is that Kafka does way more than just transporting messages but it is the distributed commit log on which the entire data flow in an organization can be based on. Its performance is very much dependent on the extent

to which it is well embedded in the overall system architecture and its constraints are properly handled.

## 5.2 Innovative Application Scenario

One of the digital revolutions where Apache Kafka is highly influential is a smart city traffic management system.

Traffic sensors, GPS devices and surveillance cameras continuously send various pieces of data that are stored in Kafka topics. Traffic patterns analysis, congestion detection, and bottlenecks prediction is some of the tasks performed by real-time processing systems like Apache Flink. As a result of these actions, traffic signals can be changed dynamically by automated systems, route alternatives can be sent to drivers, and emergency services can be notified in the case of an accident.

Kafka serves as the spine of the system that helps to ingest and distribute the streams of high-velocity data among different subsystems such as analytics engines, dashboards, and control systems. Its fault tolerance feature supports business continuity even when some of the components have failed, and this is especially crucial for urban infrastructure.

It is Kafka's ability to enable real-time, data-driven decision-making in complex, distributed environments that the above example demonstrates, thus taking it well beyond its traditional applications.

## 6. Conclusion

Apache Kafka is truly a major technology for distributed systems and, in particular, for those using the internet. In fact, its ability to process huge volumes of data quickly and in real-time, its high level of fault-tolerance and easy scalability make it a perfect partner for data-heavy applications these days. Its architecture composed of multiple servers, data partitioning, replicating, and different user groups enabled, Kafka has a toolkit for the most significant problems like reliability, 24/7, and decoupled systems.

This research on Kafka exposed the utilization of basic distributed systems ideas such as the ordering of events, leader election, and replication in order to achieve a system that is both robust and high-performing. Apart from that, actual case studies were reviewed to demonstrate how Kafka can be applied in various environments such as analytics, microservices, and financial systems. Furthermore, issues like operational complexity and the limitation of ordering only within a partition highlight the importance of careful system design and management.

Major aspects like release of the Kafka feature that no longer requires Apache ZooKeeper and the growth of deployment of cloud-native are evidence that Kafka is transforming and enhancing itself to support the industry trends.

In conclusion Apache Kafka goes way beyond being just another message passing system; it can be thought of as a distributed platform for event streaming which is essential

for real-time, scalable, and resilient applications on the web. Its ongoing improvements and vast usage show its importance in the world of distributed computing.

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